SOLVING POISSON EQUATION

Spectral Method

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Objective

Solve the 3D Poisson equation using spectral methods:

$$\nabla^2 V = \sin(x + y + z), \quad \text{for } x, y, z \in [0, 2\pi]$$

Conditions

- Periodic boundary conditions in all directions
- Computational domain: cube $[0, 2\pi]^3$

Goal

- Solve the equation using a spectral (Fourier-based) method
- Visualize the solution using color plots of 2D slices through the cube

What is Spectral Method

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Spectral methods solve differential equations by transforming them into the frequency domain using series expansions such as the Fourier series.

Periodic Functions & Fourier Series

A continuously differentiable periodic function $f(x)$ with period L satisfies:

$$f(x \pm L) = f(x)$$

It can be expanded as:

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{ik_n x}, \quad \text{where } k_n = \frac{2\pi n}{L}$$

The Fourier coefficients c_k are given by:

$$c_k = \sum_{n=0}^{N-1} f_n \cdot e^{-ik\frac{2\pi}{N}n}, \quad \text{for } k = 0, 1, \dots, N-1$$

Spectral methods exploit the ease of differentiation in the Fourier domain, turning differential operations into simple algebraic ones.

Simple Example using FFT

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We solve:

$$u''(x) = \sin(4x), \quad x \in [0, 2\pi], \text{ with periodic boundary conditions}$$

The function $u(x)$ is expanded in a Fourier series:

$$u(x) = \sum_{k=-\infty}^{\infty} \hat{c}_k e^{ikx} \qquad u''(x) = \sum_{k=-\infty}^{\infty} -k^2 c_k e^{ikx}$$

We express the right-hand side as a Fourier series:

$$\sin(4x) = \frac{1}{2i}\left(e^{i4x} - e^{-i4x}\right)$$

So the non-zero Fourier coefficients of $\sin(4x)$ are:

$$\hat{f}_4 = \frac{1}{2i}, \quad \hat{f}_{-4} = -\frac{1}{2i}$$

Simple Example using FFT

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From the spectral method, we equate coefficients:

$$-k^2 c_k = \hat{f}_k \quad \Rightarrow \quad c_k = \frac{-\hat{f}_k}{k^2}$$

Thus, only the $k = \pm 4$ modes are non-zero:

$$c_4 = \frac{-1}{2i \cdot 16} = \frac{-1}{32i}, \quad c_{-4} = \frac{1}{32i}$$

We write the solution:

$$u(x) = c_4 e^{i4x} + c_{-4} e^{-i4x} = \frac{-1}{32i} e^{i4x} + \frac{1}{32i} e^{-i4x}$$

Using the identity:

$$e^{i4x} - e^{-i4x} = 2i \sin(4x)$$

We simplify:

$$u(x) = \frac{-1}{32i} \left(e^{i4x} - e^{-i4x}\right) = \frac{-1}{16} \sin(4x)$$

$$\boxed{u(x) = -\frac{1}{16} \sin(4x)}$$

This is the exact solution obtained via the spectral method.

Simple Example using FFT

We solve:

$$u''(x) = \sin(4x), \quad x \in [0, 2\pi], \text{ with periodic boundary conditions}$$

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Steps:

1. **Collocation Points:** Discretize $[0, 2\pi]$ into $N = 1001$ equispaced points.
2. **Define RHS:** $f(x) = \sin(4x)$ evaluated on the grid points
3. **Forward FFT:** Apply `fft` to $f(x)$ to get its DFT: $\hat{f}_n$.
4. **Shift Spectrum:** Use `fftshift` to center frequency components in $\hat{f}_n$.

```
#create collocation points
x_arr = np.linspace(0,2*np.pi,N+1)
x_n = x_arr[:-1]

#define the source rhs term
s_rhs = np.sin(4*x_n)

#forward DFT
dft_s = fft(s_rhs)

#recentering
ft_s = fftshift(dft_s)
```

Continue...

5. **Construct Wavenumber Array:** Define $k_n = i - \frac{N-1}{2}$ for $i = 0$ to $N-1$.

6. **Solve in Fourier Space:**

$$c_n = \begin{cases} -\hat{f}_n / k_n^2, & k_n \neq 0 \\ 0, & k_n = 0 \end{cases}$$

7. **Inverse Shift:** Apply `ifftshift` to reorder $\hat{u}_n$ back to DFT-compatible form.

8. **Inverse FFT:** Use `ifft` on c_n to obtain $u(x)$ in physical space.

```python
#defining the k array
k_arr = np.zeros(N)
for i in range (0,N):
    k_arr[i]=i-(N-1)/2

#define the Fourier coefficients
c_n = np.zeros(N)*(1.0+0.0j)
for i in range(0,N):
    if (i==(N-1)/2):
        c_n[i]=0.0
    else:
        c_n[i]=-ft_s[i]/(k_arr[i]**2)

#rearrange order to get it in dft ordering
dft_c = ifftshift(c_n)

#inverse fourier transform
f = ifft(dft_c)
```

Output:

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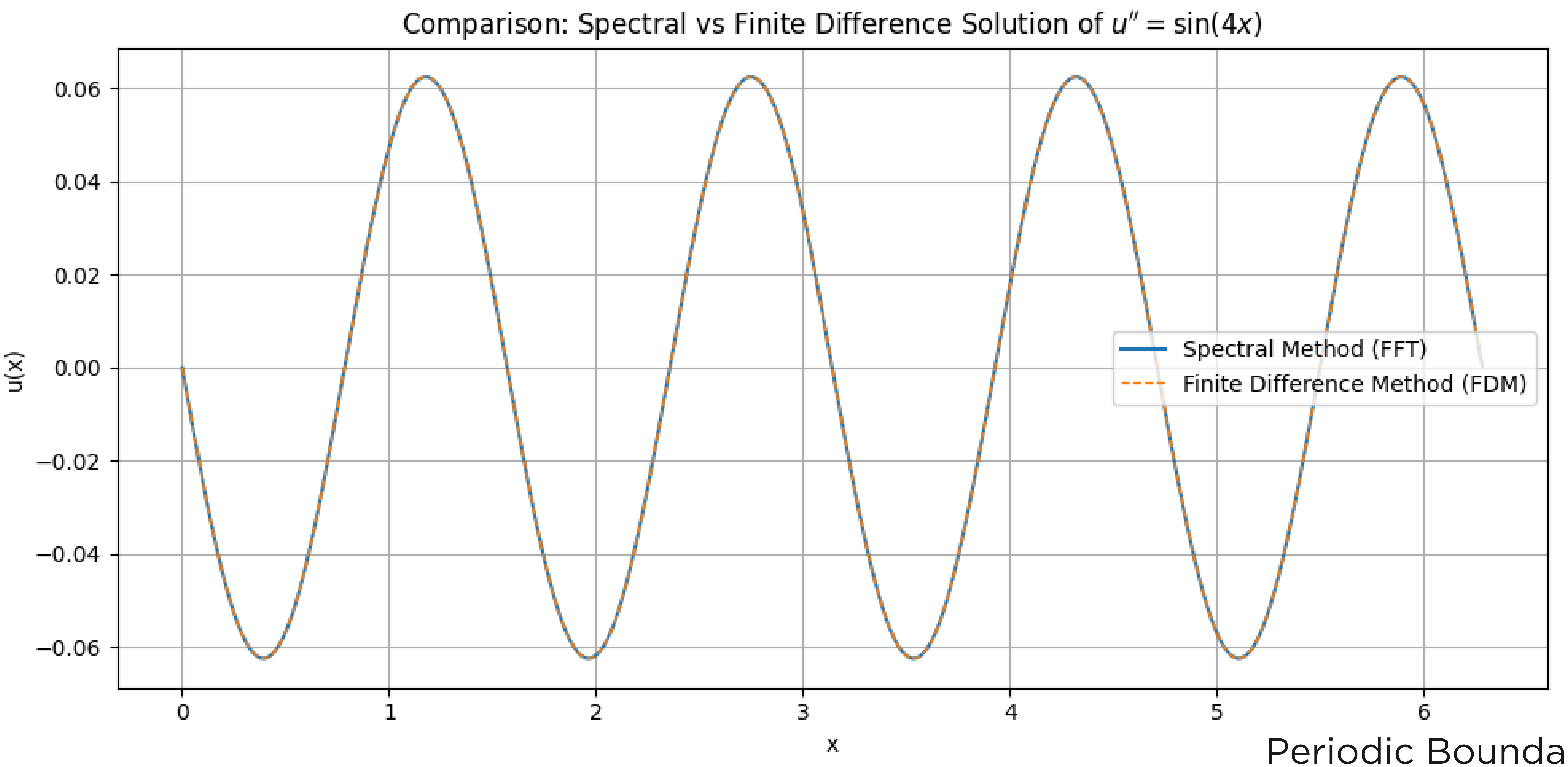

Periodic Boundary Conditions

- u(0)=u(2π)
- u′(0)=u′(2π):

Flow Chart

$u''(x) = g(x)$

$$
\begin{tikzcd}
\text{g(x) in discrite form} \arrow[r, "\text{fft}"] & \text{Fourier transform of g : } \hat{f}_k \arrow[d, "c_k = \frac{-\hat{f}_k}{k^2}"] \\
\text{Inverse fourier transform} & c_k \arrow[l, "\text{ifft}"']
\end{tikzcd}
$$

Why Do We Need fftshift ?

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`fftshift` is used to **reorder the frequency components** so that the zero-frequency mode $\hat{u}_0$ is centered in the array. This makes the frequency spectrum symmetric around $k = 0$, which is convenient for both visualization and differentiation in spectral methods.

$$\begin{bmatrix} \text{Before:} & \hat{u}_0 & \hat{u}_1 & \cdots & \hat{u}_{N/2-1} & \hat{u}_{-N/2} & \cdots & \hat{u}_{-1} \\ \text{After:} & \hat{u}_{-N/2} & \cdots & \hat{u}_{-1} & \hat{u}_0 & \hat{u}_1 & \cdots & \hat{u}_{N/2-1} \end{bmatrix}$$

Poisson Equation

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$$\nabla^2 V(x, y, z) = \sin(x + y + z)$$

inside a cube $[0, 2\pi]^3$ with periodic boundary conditions using the spectral method.

1. **Collocation Points:** Discretize each spatial dimension $x, y, z \in [0, 2\pi]$ using $N = 101$ equispaced grid points.

2. **RHS Evaluation:** Define the right-hand side function:

$$f(x, y, z) = \sin(x + y + z)$$

and evaluate it at all grid points.

3. **3D FFT:** Apply the 3D Fast Fourier Transform (FFT) to $f(x, y, z)$ to obtain its spectral representation $\hat{f}(k_x, k_y, k_z)$

```
# Step 1: Collocation Points
N = 101
L = 2 * np.pi
x = np.linspace(0, L, N, endpoint=False)
y = np.linspace(0, L, N, endpoint=False)
z = np.linspace(0, L, N, endpoint=False)
X, Y, Z = np.meshgrid(x, y, z, indexing='

# Step 2: RHS f(x, y, z)
f = np.sin(X + Y + Z)

# Step 3: Forward 3D FFT
f_hat = fftn(f)
```

Continue...

4. **Construct Wavenumber Arrays:** Define the wavenumber vectors k_x, k_y, k_z using the FFT frequencies.

5. **Solve in Fourier Space:** The solution in Fourier space is given by:

$$\hat{V}(k_x, k_y, k_z) = \begin{cases} -\dfrac{\hat{f}(k_x, k_y, k_z)}{k_x^2 + k_y^2 + k_z^2}, & \text{if } k_x^2 + k_y^2 + k_z^2 \neq 0 \\ 0, & \text{otherwise} \end{cases}$$

```
# Step 4: Wavenumber Arrays
k = fftfreq(N, d=(L/N)) * 2 * np.pi
kx, ky, kz = np.meshgrid(k, k, k, indexing='ij')

# Step 5: Solve in Fourier Space
k2 = kx**2 + ky**2 + kz**2
k2[0, 0, 0] = 1  # avoid division by zero
V_hat = -f_hat / k2
V_hat[0, 0, 0] = 0  # set mean to zero

# Step 6: Inverse FFT
V = np.real(ifftn(V_hat))
```

6. **Inverse FFT:** Apply the inverse FFT to $\hat{V}(k_x, k_y, k_z)$ to get the solution $V(x, y, z)$ in physical space.

Continue...

8. **2D Plots:** Visualize 2D slices of $V(x, y, z)$ along the mid-planes $x = \pi$, $y = \pi$, and $z = \pi$.

```
# Step 7: Plot 2D slices
slice_index = N // 2

plt.figure(figsize=(18, 5))

plt.subplot(1, 3, 1)
plt.imshow(V[:, :, slice_index], extent=[0, L, 0, L], origin='lower', cmap='viridis')
plt.title('Slice at z = π')
plt.colorbar()

plt.subplot(1, 3, 2)
plt.imshow(V[:, slice_index, :], extent=[0, L, 0, L], origin='lower', cmap='plasma')
plt.title('Slice at y = π')
plt.colorbar()

plt.subplot(1, 3, 3)
plt.imshow(V[slice_index, :, :], extent=[0, L, 0, L], origin='lower', cmap='coolwarm')
plt.title('Slice at x = π')
plt.colorbar()

plt.tight_layout()
plt.show()
```

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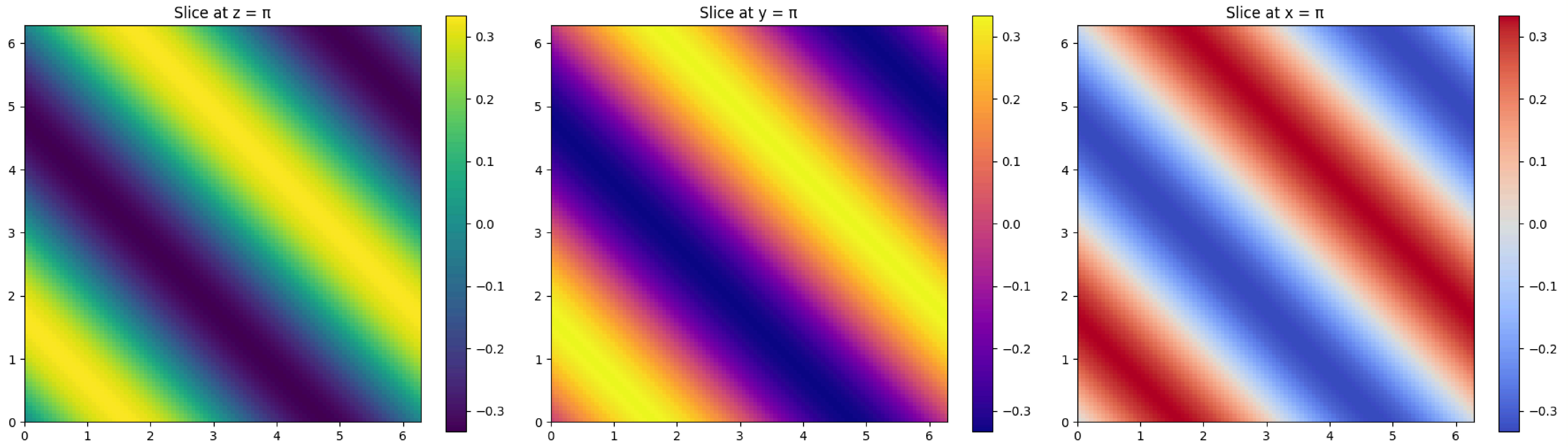
Output:



Analytical Solution of Poisson's Equation

Problem: Solve the Poisson equation inside a periodic cube:

$$\nabla^2 V = \sin(x+y+z), \quad (x,y,z) \in [0, 2\pi]^3$$

Approach: Use the spectral method with periodic boundary conditions.

- Fourier expansion of source term:

$$\sin(x+y+z) = \frac{1}{2i}\left(e^{i(x+y+z)} - e^{-i(x+y+z)}\right)$$

- Spectral space relationship:

$$-|\mathbf{k}|^2 \hat{V}(\mathbf{k}) = \hat{f}(\mathbf{k}) \Rightarrow \hat{V}(\mathbf{k}) = -\frac{\hat{f}(\mathbf{k})}{|\mathbf{k}|^2}$$

- For $\mathbf{k} = (\pm 1, \pm 1, \pm 1)$, $|\mathbf{k}|^2 = 3$:

$$\hat{V}(1,1,1) = -\frac{1}{6i}, \quad \hat{V}(-1,-1,-1) = \frac{1}{6i}$$

- Inverse Fourier transform gives:

$$V(x,y,z) = -\frac{1}{3}\sin(x+y+z)$$

Thank you

This is all from group #8

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